**Code No: 117HN** **R13**

**JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD**
**B. Tech IV Year I Semester Examinations, September - 2021**
**SOFT COMPUTING**
**(Common to CSE, IT)**

**Time: 3 Hours** **Max. Marks: 75**

**Answer any Five Questions**
**All Questions Carry Equal Marks**
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1.a) Write about generate and test algorithm.
b) List the characteristics of artificial intelligence techniques. [7+8]

2. How predicate logic is used to represent knowledge and in question answering? Quote suitable examples. [15]

3. Illustrate Best first search Algorithm with suitable example. [15]

4. Perform analysis of energy function for bidirectional associative memory. [15]

5.a) Construct an ART1 network for clustering four input vectors with low vigilance parameter of 0.4 into three clusters. The four input vectors are [0 0 0 1], [0 1 0 1], [0 0 1 1] and [1 0 0 0]. Assume the necessary parameters needed.
b) Briefly discuss the basic functionality of ART Networks. [7+8]

6.a) Differentiate between Cauchy machine and Gaussian machine.
b) Explain the general structure of full Counter Propagation Net. [7+8]

7.a) Discuss fuzzy set application in engineering sector.
b) List the properties of fuzzy sets. [7+8]

8.a) Explain the importance of multi person decision making.
b) Discuss the functionality of intelligent agent for Internet search. [7+8]

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